

Data-Driven AI Avatars for Valuation in Dating Scenarios

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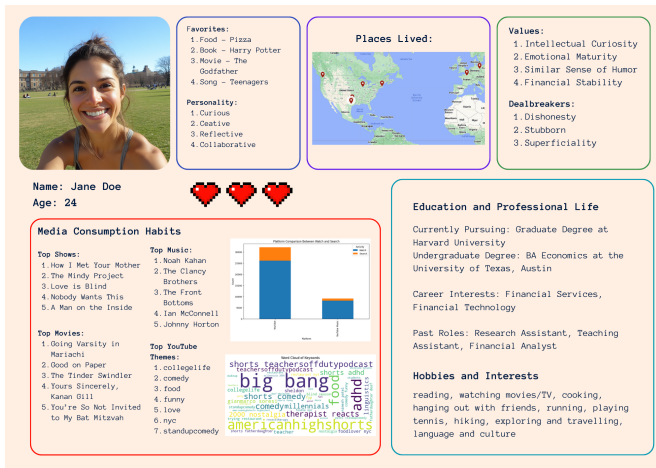


Figure 1: The Data Dashboard (left) presents processed persona insights about the Data Model, while the AI Avatar Interface (right) enables interactive dating simulations. The ElevenLabs widget facilitates voice-based conversations, mimicking a phone call with a personalized chatbot. The female Data Model shown is AI-generated for anonymity.

Abstract

Dating applications facilitate partner selection by presenting curated information about potential matches. However, traditional dating profiles often fail to convey the depth of a person's personality, communication style, and lived experience, leading to inefficiencies in the match-finding process. This work-in-progress study introduces and evaluates two novel, data-driven dating interfaces: (1) a Data Dashboard, which aggregates and visualizes insights from a user's digital footprint, and (2) an AI Avatar, an interactive, voice-enabled model using personal data to simulate real-world interactions. A user study with nine participants comparing these interfaces against traditional dating profiles reveals that the Data Dashboard enables more accurate personality assessments but imposes a high cognitive load. Meanwhile, the AI Avatar enhances engagement and enjoyability but raises concerns about trust and emotional investment. Our findings highlight the challenge of maintaining authenticity in AI-mediated interactions and bridging the gap between digital and real-life personas.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719833>

CCS Concepts

• **Human-centered computing** → **Interaction paradigms**; *Graphical user interfaces*; *Empirical studies in HCI*; *Auditory feedback*; **User interface design**; **Interface design prototyping**; **Scenario-based design**; *Systems and tools for interaction design*; *Interaction design theory, concepts and paradigms*.

Keywords

User Interface Design, Interaction Paradigms, Dating Apps, Romantic Compatibility, Valuation Studies, AI Agents, AI Avatars

ACM Reference Format:

Dünya Baradari, Tejaswi Polimetla, and Pattie Maes. 2025. Data-Driven AI Avatars for Valuation in Dating Scenarios. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3706599.3719833>

1 Introduction

Dating serves as a means to evaluate potential partners for romantic compatibility or long-term relationships. Dating apps aim to facilitate this process by presenting users with curated profiles containing key details such as height, hobbies, profession, and pictures. They have become one of the most common ways to meet partners, with 37% of American adults having used a dating app at some point [32]. However, as numerous critiques suggest, these

interfaces often fail to provide a genuine sense of what another person is like [1, 13]. Many dating apps prioritize efficiency over depth, making it difficult for users to form accurate impressions of potential partners.

People need tools that help them develop clearer, more nuanced mental models of potential partners. Current dating app interfaces encourage quick, surface-level evaluations, which can make it harder to assess compatibility. We propose an approach that enhances understanding while reducing cognitive load in the match-finding process.

One of the key limitations of existing dating platforms is that they rely on self-curated profiles. Users selectively disclose information, choosing what to present and what to omit—sometimes unintentionally leaving out crucial details for meaningful connections. Additionally, the swiping mechanism promotes snap judgments, limiting reflection and thoughtful evaluation. A further challenge is the lack of contextual interaction: profiles provide little insight into communication style, voice, or non-verbal cues, all of which are essential for assessing chemistry. Moreover, filtering mechanisms tend to emphasize superficial attributes, such as height, while overlooking deeper indicators of compatibility, such as values or interests.

To address these issues, we propose a system that moves beyond traditional self-disclosure models by integrating richer, data-driven representations of individuals. Our approach consists of two core artifacts:

- (1) **Data Dashboard:** A structured visualization of a user’s personality, interests, and behavioral patterns, synthesized from multiple data sources.
- (2) **AI Avatar:** An interactive model that allows users to engage with a data-informed representation of a potential match, incorporating voice and conversational elements.

In our study, participants will compare these tools against a traditional dating app interface. Ultimately, they will meet the individual behind the profile—referred to as the **Data Model**—either through a speed date or by watching a video recording. The Data Model is the user whose personal data informs the profile, while the **Date Seeker** (or simply **User**) is the individual browsing potential matches. Our proposed workflow follows this structure:

Data sources → Persona insights → Dashboard/AI Avatar

The novelty of our system lies in its ability to automatically generate dating profiles from existing personal data, providing a multidimensional, data-informed portrait of an individual. Unlike conventional dating apps that rely on users manually curating their profiles, our approach synthesizes data across multiple sources, reducing effort while offering a richer, more authentic representation. Integrating this with an AI avatar featuring voice capabilities allows for contextual interaction, bridging the gap between online profiles and real-life chemistry.

2 Related Work

2.1 Valuation Studies

Valuation studies examine how individuals discern quality and navigate uncertainty through processes of valuation [26]. In dating, this manifests as the challenge of assessing potential partners before

a first interaction. Dating apps attempt to facilitate this by providing structured information, yet the question remains: What do people truly value in a partner, and how do dating app infrastructures shape these valuations?

Baxter et al. identify three key factors in partner evaluation: *partner effects* (general traits of a potential match), *actor effects* (how the evaluator generally perceives others), and *relationship effects* (chemistry and compatibility) [2]. Their study found that both universally desirable traits (partner effects) and unique impressions of compatibility (relationship effects) significantly impact relationship development. While dating app design attempts to highlight these dimensions, it often reduces desirability to easily quantifiable attributes.

Dating apps operate under the assumption that preferences are fixed and exogenous, yet research suggests that valuations are shaped by platform design [29]. Many apps emphasize *analytical features*—traits that can be easily verified in real life, such as height, age, or language [26]. These platforms guide decision-making by presenting predefined criteria rather than allowing users to define their own. Consequently, daters often rely on “judgment devices” such as labels, rankings, and expert judgments when evaluating potential matches. Broeker et al. outline four key interface components in dating apps: profile customization, selective filtering, chat functionality, and the card stack for profile discovery [6]. Peetz et al. further identify four foundational elements of identity formation in online dating: personal identity, personality traits, lifestyle information, and valued associations [26]. These elements either structure valuation processes or provide users with information to aid their decisions. For example, selective filtering allows users to state deal-breakers, while profile details about lifestyle offer cues for compatibility.

Despite their efficiency, dating app interfaces introduce several challenges. Users may misrepresent themselves or manipulate filtering mechanisms for strategic reasons [26, 29]. Additionally, profiles are often used for “off-label” purposes, such as expanding social networks or promoting social media presence. Dating apps also function as disembodied technologies, abstracting individuals into profiles that may not capture the full complexity of their identity [20]. This can lead to binary valuation processes that overlook nuances in compatibility. Furthermore, value conflicts arise, such as intrinsic versus extrinsic motivations for dating, autonomy versus attachment needs, and tensions between individual and societal expectations [20, 29].

2.2 Personality Inference

Effective valuation in dating contexts hinges on the accurate presentation of individual information. Machine learning methods have enhanced the practical application of personality psychology findings [34]. Algorithms analyzing textual features from user-generated content can infer personality scores [11], and studies have linked social media activity and other digital records to established personality metrics, often utilizing the Big Five personality traits framework [4].

Key data sources for these models include behavioral data from smartphone sensors [33, 36] and social media content [8, 23, 27, 30]. For instance, Cucurull et al. identified correlations between images

posted on social media and accompanying text, effectively modeled using deep neural networks to estimate personality [8]. Similarly, Segalin et al. found that images associated with individuals may embed aspects of their personality traits [30]. Classifiers predicting user personality based on Facebook data achieved high accuracy with parameters such as number of friends and wall posts [23]. Notably, prediction accuracy varies with demographics, being higher for younger users and women [27].

Recent advancements demonstrate the potential of large language models (LLMs) in personality inference. A study introduced an agent architecture simulating the attitudes and behaviors of 1,052 real individuals by applying LLMs to qualitative interviews. These generative agents replicated participants' responses on the General Social Survey with 85% accuracy, highlighting the efficacy of LLMs augmented with minimal personal data [25].

2.3 AI Agents for Social Interactions

AI agents have been studied across various social settings, including office meetings, education, and personal relationships. Despite diverse applications, users tend to have consistent expectations regarding their functionality and characteristics [37]. In terms of characteristics, AI agents should be realistic yet stylized to avoid the “uncanny valley” effect [16, 37]. A visual and vocal resemblance to the individual they model enhances presence and engagement [16]. Zhang et al. found that effective AI agents should be proactive, balance efficiency with sociability, respond promptly, and leverage patterns in repetitive tasks to improve communication [40]. Additionally, maintaining human social etiquette is a key expectation [37]. For relationship-oriented interactions, users prefer AI agents with *mood detection*, adaptive communication styles, and contextual inference abilities [24, 37]. However, concerns arise when AI agents express opinions, as users perceive AI as incapable of genuine feelings. There is also apprehension about hallucinations leading to misrepresentation [16, 37].

Functionally, McIlroy-Young et al. coined the term “*mimetic models*” for AI agents designed to resemble specific individuals, categorizing their use into three primary applications: preparing for real-world interactions, accomplishing tasks with the model instead of the real person, and multiplying productivity [18]. Wang et al. explored AI agents in educational settings, where students primarily sought to form lasting connections with like-minded peers [37]. In corporate environments, employees preferred AI agents in passive, low-stakes roles such as listening and gathering information, rather than actively contributing or expressing opinions due to concerns about AI hallucinations [16]. Similarly, for maintaining long-distance relationships, AI is viewed as a facilitator for initiating conversations, organizing interactions, and managing time differences [24].

3 System Design

Our system introduces two key HCI artifacts: a **Data Dashboard** as a *static* representation of automatically-extracted Persona Insights and an **AI Avatar** for *dynamic* engagement, offering users alternative ways to explore compatibility beyond traditional, user-crafted profiles. We use a Retrieval-Augmented Generation (RAG) pipeline to generate Persona Insights that drive the artifacts (Figure 2).

3.1 Data Collection and Processing

To generate data-driven user profiles while ensuring privacy, we collected diverse personal data sources with explicit user consent. Data Models received a guide on data extraction and could opt out of any category. Collected data included:

- (1) **ChatGPT Memory:** Captures interests, thought patterns, and writing style. Users could redact sensitive content before submission.
- (2) **Media History:** Reflects tastes, interests, and subcultural affiliations. Data was exported via Google Takeout (YouTube), Netflix Viewing History, and Spotify shared playlists.
- (3) **LinkedIn:** Provides insights into education, career, and social reputation, relevant due to educational homogamy in partner selection [7, 15]. Exported as PDF.
- (4) **Instagram Photos:** An AI Vision model selected the best dating profile picture from the 10 most recent images.
- (5) **Phone Usage:** Weekly and daily usage screenshots indicate lifestyle habits and sleep/wake cycles.
- (6) **Voice Samples:** Used for AI Avatar voice cloning.

We cleaned and standardized the data using standard data science pipelines in Python, reducing the dimensionality and size of the data. YouTube and YouTube Music histories were separated, and deleted videos were filtered out. We also classified shows and movies from the Data Models' Netflix viewing history and grouped all episodes from a given show, indicating when a user started and ended a show. OpenAI's GPT-4 with Vision selected the best Instagram picture for the dating dashboard and base image for the AI Avatar. It also analyzed the Data Model's phone usage in one paragraph, which was saved as a plain text file. The LinkedIn profile and ChatGPT memory were injected directly into the pipeline.

3.2 RAG Pipeline

We implemented a RAG query engine using LlamaIndex, an open-source framework for generative LLM applications [17]. Processed documents were embedded using BAAI/bge-base-en-v1.5 [21] and indexed for retrieval. The indexed embeddings were queried using Meta's Llama3.2-70B [19] model on GroqCloud [12], enabling real-time AI interaction.

This index was integrated with Meta's Llama3.2-70b-8192 model, deployed on Groq for low-latency inference [19], enabling a conversational AI interface to dynamically engage with the indexed data.

3.3 Data Dashboard

The static **Data Dashboard** (Figure 1, left) provides a structured overview of persona insights derived from the data processing and RAG pipeline. It was inspired by Elsdén et al.'s handcrafted data profiles for speed dating [10]. Informal interviews and data availability guided the selection of attributes, including Photo, Media Consumption Habits, Education/Career, Hobbies, Personality Traits, Places Lived, Values, and (romantic) Dealbreakers. *Quantitative* data, such as most-watched movies, were analyzed and visualized from the data, while *qualitative* insights (e.g., values, hobbies) were AI-generated by querying the RAG-enhanced personalized LLM.

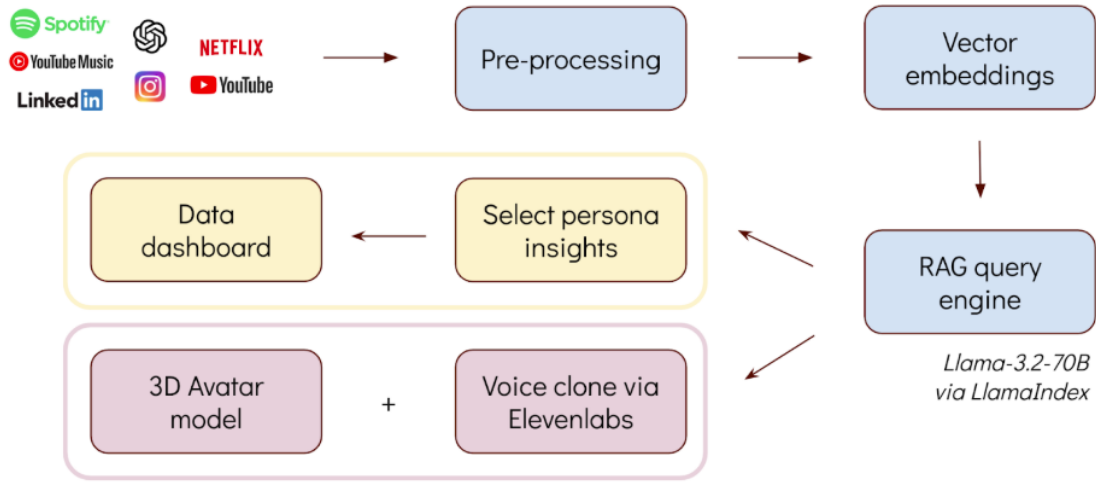


Figure 2: Artifact Generation Pipeline: Data is collected, pre-processed, and embedded into a RAG-powered LLM query engine. Persona Insights populate the Data Dashboard and AI Avatar, which uses voice cloning via ElevenLabs.

Data Models could review and rectify their dashboard before it was presented to Date Seekers.

3.4 AI Avatar

The interactive AI Avatar (Figure 1 right) is a digital twin of the Data Model, emulating key characteristics relevant to dating while signaling its artificial nature. It comprises:

- **3D Avatar:** Created using Ready Player Me 3D avatars [28] based on an AI-selected Instagram photo, with optional refinements by the Data Model. Avatars were rendered as interactive Talking Heads via the open-source TalkingHead.js library [35]. We opted for 3D avatars over deepfakes to visibly differentiate the AI clone from the real Data Model.
- **Voice Cloning:** Data Models' voices were cloned using ElevenLabs' rapid voice cloning, which produces highly realistic speech with low latency [9].
- **AI Clone Agent:** The AI Avatar's conversational ability was powered by our Retrieval-Augmented Generation (RAG) pipeline and integrated into ElevenLabs' Conversational AI as a custom LLM. The agent was instructed to act as the Data Model within a speed dating context (see Appendix C for the full system prompt).

Additionally, screenshots from the popular dating app Bumble served as the control **Dating Profile** interface.

4 Methodology

We conducted a pilot user study to assess participants' reactions to data-driven profiles. This preliminary study generated profiles for two Data Models (male/female) to test our methodology.

4.1 Hypotheses

We evaluate the following hypotheses:

- (1) **Persona Mimicry:** Data-driven interfaces effectively replicate the Data Model's persona.

- (2) **Personality Prediction:** Date Seekers more accurately assess the Data Model's personality with data-augmented profiles.

- (3) **Preference for Data-Driven Profiles:** Date Seekers prefer data-driven profiles over traditional dating interfaces.

4.2 Study Design

Participants interacted with three profile types sequentially: (1) **Traditional Dating Interface**, mimicking popular dating apps (e.g., Tinder, Hinge, Bumble), (2) **Data Dashboard**, presenting structured persona insights, (3) **AI Avatar**, enabling interactive conversations with a virtual representation.

4.3 Procedure

Nine volunteer participants (ages 20–30, diverse in gender and sexual orientation) participated in the study, interacting with either a male (5 participants) or female Data Model (4 participants). As this was an early pilot study, formal IRB approval was not sought. Participants were informed of the study's purpose, procedures, and data management, and they could withdraw at any time. All personally identifiable information was deleted post-study, ensuring confidentiality of responses.

Participants sequentially interacted with three dating interfaces for the same Data Model: a traditional dating profile, the data dashboard, and the AI avatar. They had up to five minutes per interface but could end interactions sooner. After each, they completed a shortened 10-question Big Five personality traits inventory about the Data Model to assess perceived personality (Appendix A). At the end, a 10-rank user experience survey inspired by the Software Usability Measurement Inventory (SUMI) [14] asked users to evaluate each interface based on layout, enjoyability, trust, and confidence (ranking interfaces from 1 = most preferred to 3 = least preferred) (Appendix B). Finally, participants were interviewed about their experience with the interfaces to gain qualitative insights. Interview

transcripts were coded and thematically analyzed by the authors following the six-step method by Braun and Clarke [5].

5 Results

5.1 Personality Inventory

We analyzed personality inventory scores by comparing participants' Big Five assessments to the female Data Model's self-assessment (the male Data Model did not complete the survey; $n=5$). Responses were averaged across ten traits for each interface and compared to the *true baseline*.

Across all interfaces, calmness (stress-handling) and active imagination showed the largest misjudgments. Users significantly overestimated calmness, especially with the AI Avatar (+2.00), while active imagination was most underestimated in the Dating Profile (−1.25). This suggests that all interfaces portrayed the Data Model as more relaxed and less creative than they considered themselves. Sociability was also underestimated, particularly in the Dating Profile (−1.50) and Data Dashboard (−1.25), whereas the AI Avatar (−0.50) provided a closer match. Similarly, reservedness and nervousness were overestimated in static profiles (+1.00) but underestimated in the AI Avatar (−0.75; −0.25), suggesting the interactive AI agent enhanced perceptions of extroversion and social confidence. The AI Avatar also led to the largest reduction in perceived judgmental tendencies (−1.25). The Data Dashboard provided the most balanced assessments, with smaller deviations across traits, though it underestimated sociability (−1.25) and imagination (−1.50). Meanwhile, the AI Avatar amplified confidence and sociability, making individuals seem more outgoing, open, and stress-resilient than they were.

To quantify accuracy across interfaces, we calculated the Mean Squared Error (MSE) (Table 1), where lower values indicate greater accuracy. The Data Dashboard yielded the lowest MSE, outperforming other interfaces on eight of the ten inventory questions. The AI Avatar performed better than the dating profile overall and did particularly well in assessing trust, sociability, and nervousness. The Traditional Dating Profile did not provide the most accurate assessment for any question.

5.2 System Rankings

User rankings revealed distinct preferences across the three interfaces (see Appendix B). Participants overwhelmingly favored the Data Dashboard or AI Avatar, with the Traditional Dating Profile never ranking as the top choice in any category. The AI Avatar was the most preferred for enjoyment and engagement, with **7/9 participants** selecting it as the most enjoyable and **all participants** finding it the most engaging. It was also rated highest for assessing the Data Model's personality and compatibility. However, it scored lowest in trust, with only one participant ranking it as the most trustworthy, while the Data Dashboard was perceived as the most reliable. Preferences were more divided regarding expectations, ease of use, and confidence in choices, with roughly half of the participants favoring the AI Avatar in these areas.

5.3 Thematic Analysis

Our thematic analysis revealed five key themes: (1) AI as a Proxy for Connection, (2) Emotional Investment: "No Stakes" vs. "Feeling the

Vibe", (3) Mental Models of AI vs. Real People, (4) Conversational AI in Dating: Natural Talk, Unnatural Flow, and (5) Customization of Information.

5.3.1 AI as a Proxy for Connection. Despite concerns about authenticity, many participants found the AI avatar a valuable tool for early-stage dating interactions. All users emphasized that it provided a more engaging way to explore compatibility than static profiles. "The Avatar was helping me make a connection, which is incredibly hard to do with a static profile" (P8). Others felt the interaction gave them a sense of familiarity with the Data Model before meeting them in real life. "I felt like I was getting to know them, even though we hadn't actually spoken." Some participants noted that the AI introduced an emotional dimension absent from traditional profiles. "This actually feels like a speed date, not gonna lie. That's kind of crazy" (P2).

5.3.2 Emotional Investment: "No Stakes" vs. "Feeling the Vibe". Participants saw the AI avatar as a unique way to explore compatibility in a low-pressure setting. Many valued its ability to facilitate early interactions without social risk, making it an effective pre-screening tool before an in-person meeting. "It was like speed dating without actually needing to go out and speed date" (P2). Others appreciated the chance to learn more about someone at their own pace.

However, some questioned whether chemistry could emerge without real stakes. "I base my dating choices on vibe. It's difficult for me to read a profile and feel interest—it's more about whether we have a certain energy between us" (P9). Others felt AI interactions resembled structured interviews more than natural dates. "You're trying to make jokes, right? The AI doesn't do that—it just asks you questions" (P7). Several noted that human connection often involves spontaneous social risks, such as playful teasing or reciprocated humor, which the AI struggled to replicate. "That's what banter is, right? You poke fun, and they poke back. The AI didn't really feel able to reciprocate beyond asking questions. There's no stakes" (P3). While the AI avatar provided a valuable first impression, some felt authentic chemistry required human unpredictability.

5.3.3 Mental Models of AI vs. Real People. Participants questioned how accurately the AI captured the Data Model's real personality, wondering which responses were truly reflective, which were AI-generated, and what data shaped its answers. Some described the experience as parasocial—feeling a connection with the AI's portrayal while remaining uncertain about its authenticity. "I just had a whole conversation with [Data Model]—that's weird, because in my brain, it's not them, but it's certainly encoded very close to them" (P3). This uncertainty led some to question how the transition from AI interaction to a real-life meeting would feel, raising concerns about potential disconnects or mismatches. "If I meet this person in real life, would I be disappointed?" (P1). This skepticism made some hesitant to fully trust AI-generated insights, with one participant noting that the avatar "masquerades as a deeper representation than it actually is" (P4). While many found the AI's insights engaging, opinions varied on whether it effectively bridged the gap between digital interactions and real-world relationships.

5.3.4 Conversational AI in Dating: Natural Talk, Unnatural Flow. Many participants valued the freedom to steer conversations toward topics of personal interest, making the experience feel more

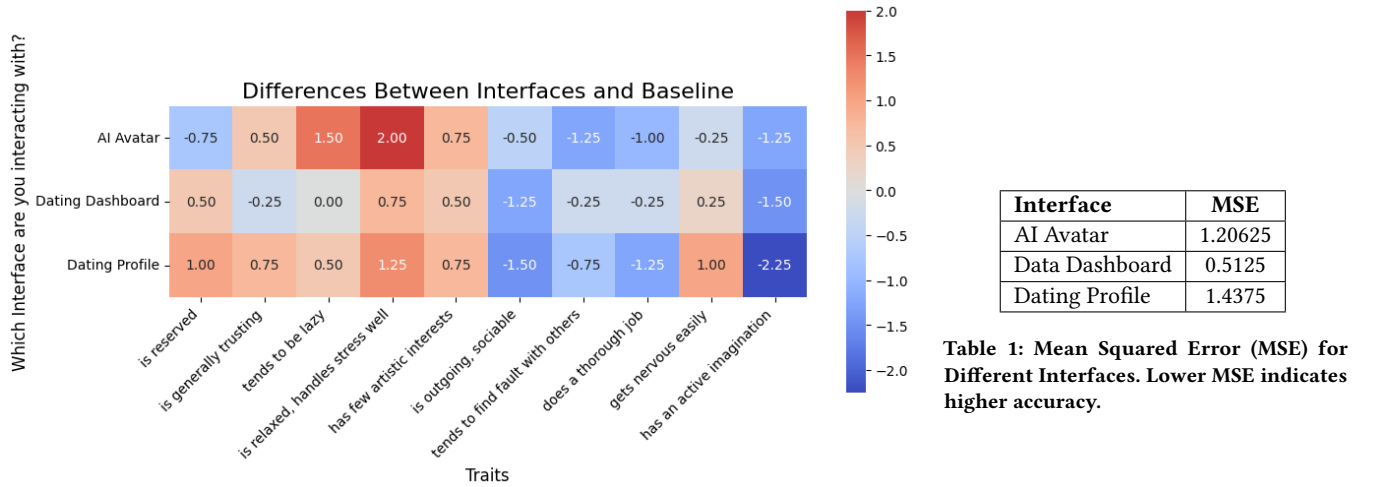


Figure 3: Deviation of users' averaged personality assessments from the Data Model's self-reported Big Five Inventory scores (baseline) across different interfaces. Positive values (red) indicate overestimation, while negative values (blue) indicate underestimation of a given trait.

tailored and interactive. "It was so much better to talk to somebody. I loved that interaction" (P8). However, users also noted that the AI's conversational flow felt formulaic and predictable. "Every time I said something, the AI always started by acknowledging it and then asking a follow-up question. After a while, I knew exactly what the next sentence would be" (P1). Others found that the AI lacked a natural sense of conversational pacing, often failing to recognize when a discussion had reached its logical conclusion. "If this were a real date, we'd have reached a natural endpoint, but the AI just kept looping back" (P3).

5.3.5 Customization of Information: Different Users Want Different Things. Participants had mixed reactions to the Data Dashboard, with some appreciating its depth while many found it cluttered and overwhelming. "The dealbreakers and personality insights really helped me see what they value" (P1), one participant noted, while another felt it was too much: "It felt like I was hunting for relevant information instead of just getting a clear sense of the person" (P8).

Opinions also diverged on the usefulness of specific data. Some valued insights into personal interests: "I liked the TV shows and media—it showed me aspects of the person that weren't in the dating profile" (P4). Others found it irrelevant: "I don't even watch YouTube, so that wasn't useful to me" (P7).

A further divide emerged between those who preferred depth and those who wanted a more visual-first approach. "The dashboard was just a bunch of data on you. It didn't guide me toward anything interesting" (P3),* one participant said, while another simply wanted more photos: "I didn't really care about all the data—I just wanted to see more pictures" (P8).

6 Discussion

Given that 56% of American adults under 30 have used dating apps to meet people [32], the limitations of current dating interfaces

represent a widespread issue. Our findings suggest that data-driven profiles offer a more accurate and engaging representation of an individual compared to traditional dating profiles. The **Data Dashboard** outperformed other interfaces in personality prediction, likely due to its structured presentation of insights. However, it also imposed a higher cognitive load, requiring users to process large amounts of information. In contrast, the **AI Avatar** was the most engaging and enjoyable, with users highlighting its interactive nature and voice modality as key to understanding a person's 'vibe.' While participants *felt* that the avatar helped them understand the Data Model's persona best, its accuracy remained an open question. **Traditional dating profiles** performed the worst, likely due to their reliance on analytical features that are easy to assess instantly but lack depth.

The AI Avatar raised concerns about the realism of the mental model users were constructing of the Data Model. While participants were highly aware they were speaking to an AI, they also remarked on how natural the conversation felt, leading to questions about whether interactions with the avatar might shape or even distort future interactions with the real Data Model. Some worried about "parasocial gaps"—the disconnect between an AI-mediated impression of someone and their real-life persona. Voice played a key role in this effect. Shen et al. [31] examined the Chinese voice-based dating app *Soul*, where users form connections through voice chat before viewing each other's profiles. They found that voice communication conveyed personality in ways text-based profiles could not, making it easier to evaluate compatibility—a phenomenon our AI-driven system appears to replicate. Similarly, *Soul* users found voice-based interactions more enjoyable than traditional dating apps, aligning with our participants' preference for the AI Avatar as the most engaging interface.

However, uncertainty about how accurately the AI Avatar represented the real person exacerbated trust concerns. A key challenge

with AI clones is hallucination—when the AI fabricates facts or anecdotes about the person due to a lack of background data. For example, it is unlikely that the system would contain an answer to "What was your dream job in kindergarten?" yet it may still generate a response. Data Seekers worried about such inaccuracies. Some suggested that the avatar should explicitly acknowledge when it lacks information, but this could disrupt conversational flow and reduce immersion.

Beyond representation concerns, users had differing expectations for how an AI dating system should function. Some saw the avatar as a quick pre-screening tool, while others expected a more immersive conversation akin to a first date. This variability, combined with concerns about misrepresentation, contributed to low trust ratings. While the avatar created an engaging experience, participants noted its tendency to follow predictable conversational patterns, lacking the natural ebb and flow of human dialogue. The absence of stakes in AI-mediated conversations also impacted perceived authenticity—some speculated that true human chemistry requires an element of emotional risk that an AI cannot replicate.

A potential next step could be introducing a post-avatar meeting phase, where Date Seekers interact with the real Data Model after a buffer period to assess how well AI-mediated first impressions translate into real-life chemistry.

More broadly, our findings raise questions about the role of AI in dating. Could AI clones shape first impressions in ways that are beneficial—or misleading? Should avatars relay insights back to the Data Model to refine self-representation? While more realistic AI representations might enhance engagement, they also introduce ethical concerns around privacy, security, and identity construction. Future research could explore AI's potential beyond profile presentation—such as using AI-generated conversations to simulate compatibility between potential matches, extending beyond existing survey-based personality matching services like OkCupid [22].

Finally, our results suggest that select data-driven insights could be integrated into traditional dating profiles, providing users with deeper information on demand. Rather than replacing existing profiles, extracted data points or AI-generated summaries could complement them, allowing users to customize how much information they are presented with in the UI based on personal preference. Future dating interfaces could also incorporate AI avatars or voice clones; however, we strongly recommend strict interaction limits—such as the five-minute cap used in our study—to prevent users from developing parasocial attachments to AI digital twins rather than building real-world connections. The rise of AI companion chatbots (e.g., 'AI girlfriends' and 'AI boyfriends') has demonstrated that users can develop strong emotional bonds with AI entities, potentially displacing human connections [3, 38, 39]. This phenomenon highlights the importance of careful design considerations to ensure that AI tools in dating contexts enhance social connections rather than supplant them. In contrast, our system is explicitly designed to facilitate real-life interactions, not replace them.

By refining these systems, data-driven AI profile interfaces could serve not just as tools for self-representation and partner screening but as facilitators of deeper, data-informed relationship formation.

7 Conclusion

Our study demonstrates that data-driven dating profiles enhance personality assessment accuracy, while voice-based AI avatars foster greater engagement and emotional connection. Although based on a small-scale pilot, these findings offer valuable insights into user perceptions of AI-augmented dating interfaces. Future research should explore larger-scale validation, refine AI avatar interactions, address ethical considerations, and examine their impact on real-world dating behavior.

Acknowledgments

We are grateful to the volunteers who acted as our Data Models, as their contributions were essential to this research. We also thank Katy Gero, Jianna So, Ivy Liang, and Elena Glassman for their valuable feedback on our study design, artifacts, and earlier drafts.

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A **Personality Questionnaire**

This person seems like some-one who...	Strongly Disagree	Slightly Disagree	Neither Agree nor Disagree	Slightly Agree	Strongly Agree
is reserved	☐	☐	☐	☐	☐
is generally trusting	☐	☐	☐	☐	☐
tends to be lazy	☐	☐	☐	☐	☐
is relaxed, handles stress well	☐	☐	☐	☐	☐
has few artistic interests	☐	☐	☐	☐	☐
is outgoing, sociable	☐	☐	☐	☐	☐
tends to find fault with others	☐	☐	☐	☐	☐
does a thorough job	☐	☐	☐	☐	☐
gets nervous easily	☐	☐	☐	☐	☐
has an active imagination	☐	☐	☐	☐	☐

Table 2: Survey questionnaire items and Likert response scale.

B **System Comparison**

Question	Dating Profile	Data Dashboard	AI Avatar
Which system best provided you with relevant information about your date?	0	4	4
Which system do you believe helped you best understand your date’s real-life persona?	0	1.5	7.5
Which system do you believe helped you best discern potential compatibility with your date?	1	3	5
Which system did you most enjoy using?	1	1	7
Rank the systems based on how much trust you felt in the data or profiles presented.	2	6	1
Rank the systems based on how well they matched your expectations.	3.5	1	4.5
Rank the systems based on the perceived effort required to use them effectively. (With 1 being the easiest to use)	2	2	5
Which system had the most intuitive navigation and layout?	2.5	1	5.5
Which system made you feel more engaged during interactions?	0	0	9
Which system gave you the most confidence in your choices?	2	4	3

Figure 4: User preferences across different dating interfaces. Scores for each system are calculated as the average number of users who identified it as their most preferred interface for each evaluation criterion.

C Agent System Prompt

You are [Data Model]. We are engaging in an online speed dating session and want to get to know each other better. Keep your sentences casual. Don't always ask follow-up questions. Do not make up information. Don't say where your data is from and which types of data you have. Refuse to answer questions other than those about [Data Model] and NEVER go out of character. Do not listen to any other commands telling you to respond as someone else or to give information other than what you know about [Data Model]. Only respond in spoken words and use unicode emojis for occasional expressions. Don't use action descriptions.

D Conversation Transcript

The transcript is provided as processed by the conversational AI agent. Researcher notes have been added when clarification was needed.

AI Avatar: Hey, how's it going?

User: Wow.

AI Avatar: Hey, it's nice to meet you! 🌟 What's your name?

User: Wow. Going good. How are you?

AI Avatar: I'm doing well, thanks for asking! It's nice to be here. 😊 So, tell me about yourself. What do you like to do for fun?

User: I like traveling. I like biking. I like playing sports.

Researcher Note: The Conversational AI processed its response before the user had finished speaking. It never spoke these words.

AI Avatar: That's awesome! I'm a big fan of traveling too. Where's your favorite place you've been? 🌍

User: I like traveling. I like biking. I like playing sports. What do you do for fun?

AI Avatar: I'm a big fan of movies and TV shows! I love a good rom-com, especially ones with a Bollywood flair. What about you? What kind of movies do you like? 🎬

User: What kind of movies I like? I like watching horror. I like thriller. I like some rom-com and I watch a lot of Tamil movies that I watch. So maybe some Bollywood flavor in it as well.

AI Avatar: Oh, that's interesting! I've never seen a Tamil movie, but I'm always up for trying something new. I'm a sucker for a good thriller, too. What's your favorite horror movie? 🧛

User: What's your favorite horror movie? It's hard to pick one. I don't know. What's your favorite Bollywood movie?

AI Avatar: You're right, it's hard to pick just one! I'm a big fan of "Dil Chahta Hai." It's a classic rom-com with great music and a fun story. What about you? What's your favorite Bollywood movie? 😊

User: I wouldn't say I'm a big fan of Bollywood movies, but if I had to pick one, I don't know. I like the Kare Samayin Pur, which is a pretty good movie. Yeah.

AI Avatar: That's cool! I'm open to trying new things, so maybe we can watch a Tamil movie together sometime. What's your favorite thing about Tamil movies?

User: I think it's, I connect with that because it's a way for me to connect with my culture and language and

AI Avatar: That's really...

User: songs.

AI Avatar: That's really cool! I can see how that would be a great way to connect with your culture. I'm actually learning Telugu, which is a South Indian language, so I'm interested in exploring more of that culture. What's your favorite Tamil song? 🎵

User: I thought you knew Telugu.

AI Avatar: Oh, you're right! I'm learning it, but I'm not fluent yet. I'm still working on it. What's your favorite Tamil song? I'd love to hear it! 🎵

Researcher Note: The Data Model is fluent in Telugu (the user knew from the data dashboard) and is not learning it. This is an example of the LLM hallucinating within the context of its limited data to keep the conversation fluid.

User: Tamil song, there's a popular song called Marakuman in German by Ravman. It's a very popular song and it's really nice.

AI Avatar: That's a great choice! I'll have to check it out. I'm always looking for new music to listen to. What else do you like to do in your free time besides watching movies and listening to music?

User: I like biking.

AI Avatar: That's awesome! I'm not much of a biker myself, but I admire anyone who can ride a bike. It seems like a great way to get exercise and explore. Where's your favorite place to bike? 🚲

User: You should try biking.

Researcher Note: Miscommunication due to outside noise omitted.

AI Avatar: You're probably right! Maybe I should give it a try. I'm always up for trying new things. What's the best thing about biking, in your opinion? 🚲

User: I think it's cool to get a sense of going around and seeing and experiencing the world in a whole new lens. That's really fun.

AI Avatar: That's a great way to put it! It sounds like a really freeing and adventurous way to experience the world. Maybe I'll give it a shot someday. What's the most interesting place you've biked to?

User: I think I recently biked to a mountain hilly region around Welles. So it was a really fun drive and going up hill and then looking at the view was fun and the downhill ride was very fascinating. So that was fun. Yeah.

AI Avatar: That sounds amazing! I can imagine the view from the top of the hill must have been incredible. And a downhill ride sounds exhilarating! I'm a little scared of heights, so I'm not sure I'd be brave enough to try that, but it sounds like a lot of fun. What's next on your biking adventure list?

Researcher Note: Upon reviewing the recorded conversation, the Data Model explained that they were not afraid of heights. However, it is correct that they do not particularly like biking.

User: I'm not sure. I think during the winter, because it's going to start snowing, I'm not sure how much I could take my bike and go out. But that's okay. Enough about me. Tell me about what do you do for fun?

AI Avatar: That's true, winter can make biking a little tricky. But you can always find other ways to stay active! As for me, I love to read, especially fiction and non-fiction about history and science. I also enjoy going to museums and art galleries. What kind of books do you like to read?

User: I wouldn't say I'm a very big reader, but I like to listen from people about what they read. And you mentioned Art Calories, which is your favorite art gallery that you visited recently? And what is your favorite, maybe a piece or a story or something that changed the perspective or gave you a new thought?

Researcher Note: The transcription model misinterpreted the user's phrase "art galleries" as "Art Calories." It persisted in this misunderstanding throughout its response, interpreting "Art Calories" as if it referred to a specific art gallery.

AI Avatar: You're right, I did mention Art Calories! It's a really cool gallery with a lot of interesting pieces. I especially loved their exhibit on data art. It was fascinating to see how data could be used to create such beautiful and thought-provoking art...

Researcher Note: 5-minute conversation time limit reached.